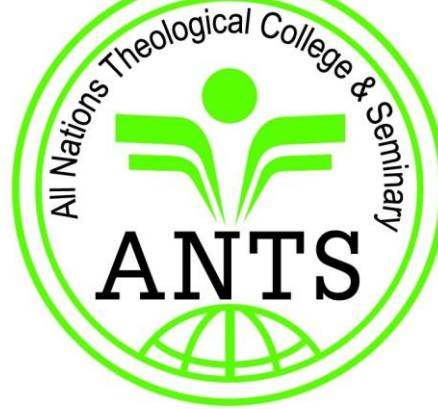




Make Disciples of All Nations

Unit 3 Operating Systems

COMPUTER LITERACY

Unit Objectives

- To define an operating system
- To describe the basic organization of computer systems and operating systems.
- To explore common types of operating systems.
- To provide a core functions of operating systems.
- To describe the services an operating system provides to users, processes, and other systems.
- To discuss the various ways of structuring an operating system.

Introduction to Operating System (OS)

- Operating system refers to the interface between the user and the hardware



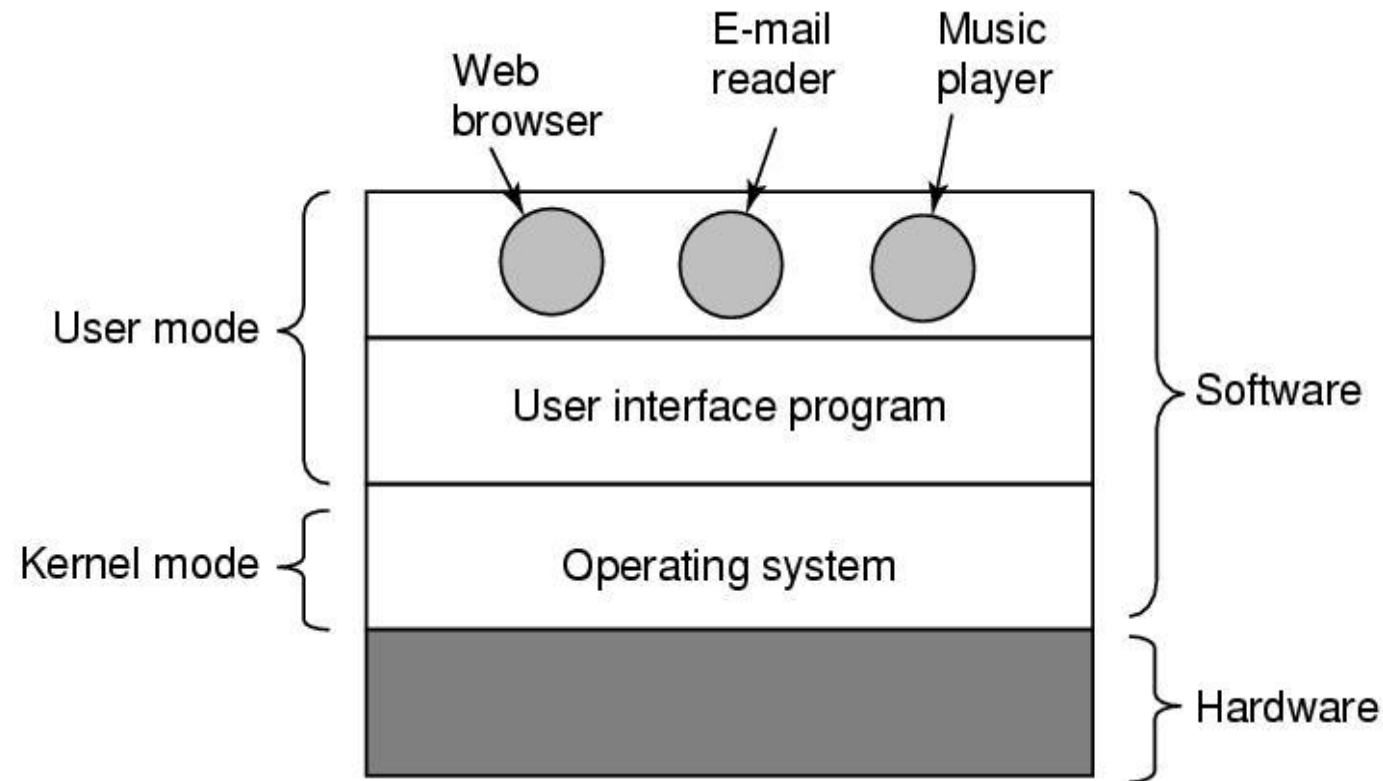
What is an operating system?

- An operating system (OS) is system software that manages computer hardware and software resources and provides common services for computer programs.
- **It acts as an intermediary between users and computer hardware.**
- A computer system has many resources (hardware and software), which may be require to complete a task.
- These resources are input/output devices, memory, file storage space, CPU etc.
- The operating system acts as a manager of the above resources and allocates them to specific programs and users, whenever necessary to perform a particular task.
- **In simple terms, an operating system is a system software that acts as an interface between the user, the application software and the computer hardware.**

Continuation...

Operating System seeks to achieve the following goals:

- Control/execute user/application programs.
- Make the computer system convenient to use.
- Ease the solving of user problems.
- Use the computer hardware in an efficient manner.
- Just like a company manager is responsible for running the affairs of a company, the OS is responsible for the smooth and efficient operation of the entire computer system



Key Functions of an Operating System

1. **Job Scheduling.** To schedule is to arrange or plan. The Operating System arranges the jobs to be carried out by the computer system. It controls the use of resources such as printers, scanners, and time slice to each user. Job scheduling is mainly done in a **First In First Out (FIFO)** basis to avoid overloading or competition for resources.
2. **Resource Control.** The processor services all the peripheral devices effectively by assigning priorities to these devices. The OS allocates **interrupt request numbers** which are used by the devices to request for attention. When two or more devices request for attention at the same time, the Operating System checks the priority of the device and allows the one with a higher interrupt number to be served first. For example, a keyboard has a higher priority than a printer and it is usually attended to first.

Key Functions of an Operating System

3. Input/Output (I/O) Handling. The Operating System controls the selection and use of input, output and storage devices in the computer. Each of those devices requires its own specific instruction files (device drivers) or control signals which are loaded by the Operating System when required. For example, during the printing process, it selects the printer to be used, prints, purges (cancels), pauses or prioritizes print jobs depending on the requirements of the user.

4. Memory Management. The Operating System allocates itself memory and ensures that each program has enough memory to execute processes. It also ensures that one process does not use the memory that has been allocated to another process. This must also be done in the most efficient manner.

Key Functions of an Operating System

5. Error Handling. The Operating System alerts the user of any errors made as one uses the computer. For example, if the start files get corrupted, the OS gives an alert to the user, and also offers the options available for solving the problem. The OS continually checks the system for any possible sources of errors and alerts the user of the same. Again, it offers the possible corrective measures. The user can enable or disable the OS to be reporting on any errors directly to the manufacturer.

6. Program Loading. For a program to be used, it has to be transferred from the secondary memory to the RAM. It is the function of the OS to load programs or subprograms. The biggest part of the OS, however, remains in the disk and is loaded to memory only when it is needed. The OS also provides a continuous sequence of processing or response to events such as saving information or printing. This is only possible if the necessary data and instructions have been loaded into the main memory.

Key Functions of an Operating System

7. File System Management. The Operating System allows the user's data and program files to be managed effectively. The OS allows the organization of files into directories and subdirectories. It also provides the environment for files to be copied, edited, and moved.

8. Disk Management. The Operating System is responsible for performing tasks such as formatting, defragmenting, compressing, disk clean up, system back up and partitioning. The Operating System is also used for copying and scanning a disk for viruses. It is also used to delete some contents of a disk in order to create more storage space. It also displays some properties such as disk volume label and bad sectors.

Key Functions of an Operating System

9. Installation and Uninstallation of Hardware and Software

10. System Security

11. User Interface (UI): Providing interaction between user and computer

12. System Settings: Configuring display, user accounts, power settings, and accessibility options.

Examples of operating systems

- Windows (Microsoft)
- Mac OS
- Linux
- Android etc..



Major Operating Systems Overview

1. Windows

- **Developer:** Microsoft Corporation **Key Characteristics:**
- Graphical User Interface (GUI) based
- Widely used in personal and business computers
- Proprietary/commercial software
- Strong compatibility with various software applications
- Regular updates and versions (Windows 10, Windows 11)

Continuation...

Advantages:

- User-friendly interface
- Extensive software compatibility
- Strong gaming support
- Comprehensive technical support

Disadvantages:

- License costs
- More vulnerable to malware
- Resource-intensive
- Less customizable

Continuation...

2. macOS

- **Developer:** Apple Inc. **Key Characteristics:**
- Unix-based operating system
- Exclusive to Apple hardware (Mac computers)
- Known for elegant design and user experience
- Strong integration with Apple ecosystem

Advantages:

- Excellent build quality and stability
- Strong security features
- Seamless integration with other Apple devices
- Professional creative software support

Disadvantages:

- Limited to Apple hardware
- Higher cost
- Less gaming support
- Limited software compatibility compared to Windows

Continuation...

3. Linux

- **Developer:** Open source community (originally by Linus Torvalds)

Key Characteristics:

- Open-source and free
- Multiple distributions (Ubuntu, Fedora, CentOS, Debian)
- Command-line interface with optional GUI
- Highly customizable and flexible

Advantages:

- Free and open source
- Highly secure and stable
- Excellent for servers and development
- Highly customizable
- Strong community support

Disadvantages:

- Steeper learning curve
- Limited commercial software support
- Less user-friendly for beginners
- Hardware compatibility issues sometimes

Mobile Operating Systems

1. Android

- **Developer:** Google (based on Linux kernel) **Key Characteristics:**
- Open-source mobile OS
- Used by multiple device manufacturers
- Google Play Store for app distribution
- Highly customizable with different launchers and interfaces

2. iOS

- **Developer:** Apple Inc. **Key Characteristics:**
- Exclusive to Apple devices (iPhone, iPad)
- Closed ecosystem with strict app approval process
- Swift/Objective-C-based app development
- Consistent user interface across devices
- Strong privacy and security focus

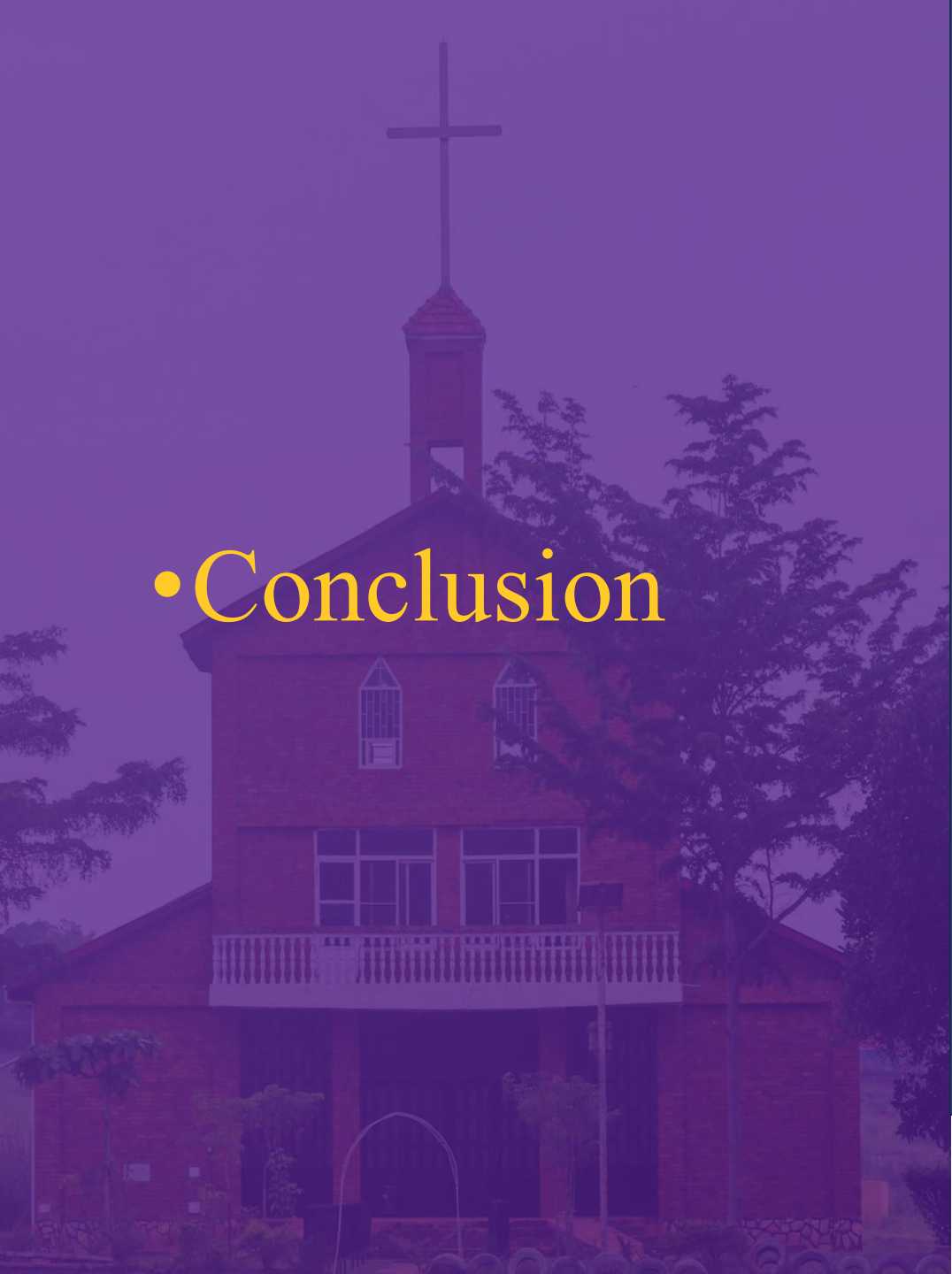
Other Mobile Operating Systems

1. HarmonyOS (Huawei):

- Developed for Huawei devices
- Microkernel-based distributed OS
- Cross-device compatibility

2. KaiOS:

- Feature phone operating system
- Based on Linux
- Supports basic smartphone features on low-cost devices



• Conclusion

- Operating systems serve as the foundation for all computer operations, managing hardware resources and providing user interfaces. Understanding the differences between Windows, macOS, and Linux helps in choosing the right platform for specific needs. Mastering file management and system settings is crucial for efficient computer use, while proper software installation and management practices ensure system stability and security.



THE END

THANK YOU!!